Ariel Gjaci

Postdoctoral Researcher | Multimodal Generative Learning | Spatial Audio | Human Motion

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Website | Linkedin | GitHub | Google Scholar

EDUCATION

Ph.D. Artificial Intelligence & Robotics
University of Genoa
Nov 2021 - May 2025 (defended)
Research collaboration with King's College London

M.Sc. Robotics Engineering
University of Genoa
2018 - 2021 | 103/110
Thesis: Culture-Aware Co-Speech Gestures using GANs

B.Sc. Biomedical Engineering
University of Genoa
2014 - 2018 | 92/110
Thesis: Activity in Hippocampal Neuronal Networks

CORE SKILLS

PROGRAMMING
Python; C/C++; R; MATLAB/Simulink; LaTeX

ML FRAMEWORKS
PyTorch; Scikit-learn; TensorFlow

METHODS
Diffusion transformers; flow matching; neural audio codecs; domain generalization

RESEARCH SYSTEMS
Linux; SLURM; multi-GPU training; Git; ROS 1

LANGUAGES

Italian (native)
English (fluent)
Albanian (native)

TRAINING

Modern Topics in Machine Learning (MaLGa); Robotics Summer School (Scuola Superiore Sant'Anna); Coursera DL/NLP courses.

PROFESSIONAL EXPERIENCE

ITALIAN INSTITUTE OF TECHNOLOGY | Postdoctoral Researcher
Feb 2025 - Present | Genoa, Italy

- Developing **FACT-RIR**: few-shot room impulse response generation from sparse acoustics, source/receiver geometry, and visual evidence.
- Designing neural audio codecs and conditional flow-matching models; studying room-level and query-dependent factors and physics-informed conditioning.
- Building reproducible HPC training and evaluation pipelines with SLURM, multi-GPU resources, controlled ablations, and multi-seed comparisons.
- Built 3D sound-source localization on room point clouds using MVDR, DAS, and MaxRE beamforming; curating spatial-audio data and writing a survey.

UNIVERSITY OF GENOA & KCL | PhD Researcher
Nov 2021 - May 2025

- Developed **SICAGE (ECCV 2026, main conference)** and ALaDiT, a diffusion-transformer model for culture-aware co-speech motion generation.
- Built **TED4C-L**: 106.45 hours from 764 speakers across four cultural groups, combining motion, audio, and text representations.
- Learned speaker-independent culture embeddings with Fishr and adversarial learning; developed semantic-similarity gesture-labeling algorithms.

AKKA TECHNOLOGIES | Software Engineer
May 2021 - Oct 2021 | Italy

- Developed Python backend software for aircraft panels and presented a working demo to stakeholders.

TEACHING & SERVICE

- Teaching assistant: Foundations of Informatics (C++), 60h; Mobile Programming (Android/Kotlin), 30h, University of Genoa.
- Co-organizer, ECCV workshop Expressive Encounters; session chair, IEEE RO-MAN 2024.
- Reviewer: AAAI, ECCV, ACM Multimedia, IROS, ICMI, and International Journal of Social Robotics.

PUBLICATIONS

Gjaci A., Sgorbissa A., Murino V. SICAGE: Speaker-Independent Culture-Aware Gesture Generation using TED4C-L Dataset. **ECCV 2026, main conference**, pp. 392-411.

Gjaci A., Schmuck V., Sgorbissa A., Celiktutan O. Exploring Cultural Cues in Multimodal Interactions. AHRI @ ACII Workshop, 2024.

Gjaci A., Recchiuto C.T., Sgorbissa A. Labeling Sentences with Symbolic and Deictic Gestures via Semantic Similarity. IEEE RO-MAN, 2024.

Gjaci A., Recchiuto C.T., Sgorbissa A. Towards Culture-Aware Co-Speech Gestures for Social Robots. Int. Journal of Social Robotics, 2022.

Gjaci A., Oneto L., Recchiuto C.T., Sgorbissa A. Culture Awareness in Intelligent Systems. AIRO Workshop, 2022.

Gjaci A., Recchiuto C.T., Sgorbissa A. A GAN-based Approach for Generating Culture-Aware Co-Speech Gestures. AIRO Workshop, 2021.